



Overestimating vs underestimating

1 An _____ is an estimate for a calculation which is greater than the exact answer. Tick **1** correct answer

- ☐ average
- ☐ impossible number
- ☐ integer
- ☒ overestimate
- ☐ underestimate

2 An _____ is an estimate for a calculation which is less than the exact answer. Tick **1** correct answer

- ☐ average
- ☐ impossible number
- ☐ integer
- ☐ overestimate
- ☒ underestimate

3 Match each calculation with a calculation that gives its overestimate. Write the correct letter in each box

a	$38.48 + 29.84$
b	$18.983 + 19.303$
c	$45.9348 + 39.84$
d	$8.548 + 2.84 + 19.203$
e	$38.48^2 + 29.84$

<i>b</i>	$\approx 20 + 20$
<i>e</i>	$\approx 40^2 + 30$
<i>d</i>	$\approx 9 + 3 + 20$
<i>a</i>	$\approx 40 + 30$
<i>c</i>	$\approx 50 + 40$



- 4 Match the following calculations which give an overestimated area. Write the correct letter in each box

a	$\approx 8 \times 8$
b	$\approx 8 + 8 + 8 + 8$
c	$\approx 5 \times 5$
d	$\approx 5 + 5 + 5 + 5$
e	$\approx 0.5 \times 0.5$
f	$\approx 0.5 + 0.5 + 0.5 + 0.5$

b	Perimeter of a square with lengths 7.89 m
a	Area of a square with lengths 7.89 m
e	Area of a square with lengths 0.498 m
f	Perimeter of a square with lengths 0.498 m
d	Perimeter of a square with lengths 4.67 m
c	Area of a square with lengths 4.67 m

- 5 Match which calculations are an overestimate, underestimate or hard to tell. Write the correct letter in each box

a	Overestimate
b	Underestimate
c	Hard to tell

c	$\frac{37.4}{4.49 + 6.41} \approx \frac{40}{5 + 6}$
a	$\frac{899}{4.40 + 6.35} \approx \frac{900}{4 + 6}$
b	$\frac{712}{8.98 + 9.45} \approx \frac{700}{10 + 10}$

- 6 Match which calculations are an overestimate, underestimate or hard to tell. Write the correct letter in each box

a	Overestimate
b	Underestimate
c	Hard to tell

a	$\frac{59 - 42}{3.67 + 1.23} \approx \frac{60 - 40}{3 + 1}$
b	$\frac{59 - 42}{3.67 + 1.23} \approx \frac{55 - 45}{4 + 2}$
c	$\frac{59 - 42}{3.67 + 1.23} \approx \frac{60 - 50}{4 + 2}$



